Positive Boundary Certificates and Logarithmic Corridors for Polynomial Vacuum Bridge Representations

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Abstract

We examine the polynomial vacuum and all-length critical bridge mechanisms in the assigned honeycomb-walk manuscript. The generic polynomial vector, its physical normalization, the complex periodic representation, and the positive bridge law are kept distinct. For the source boundary observable we develop an original-law error ledger: a positive two-external-node partition quotient, a common lower-weight kernel envelope, correlated normalization under scalar rescaling, and separate retained-pole, integration, and omitted-pole errors. The determinant identity itself is classical and already implicit in the source cofactor proof. Independently, the source lateral-tail and first-mass hypotheses imply relative preservation of the bridge mean in a corridor of radius $A(h+1)\log(h+1)$ when $cA > 11/12$. We quantify ordinary and length-biased total variation, all fixed integer length-insertion tails, and the exact normalization ratio. An abstract two-component measure shows that $11/12$ is the sharp threshold for this implication from the declared hypotheses alone; it is not a honeycomb counterexample. No cylinder exponent, bridge critical exponent, or full source proof is independently certified here. Contribution assessment: C3/5, source-specific positive boundary error budgets and a sharp conditional logarithmic-corridor transfer, without a new critical exponent or global proof certification.

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1 Physical law and proof dependencies

The assigned source is the September 26, 2026 manuscript *Polynomial vacuum representations and bridge mass for honeycomb walks* [1]. The copy used here belongs to the pinned repository tree; its identity with the captured October 8 tree is provenance, not an assertion about a live branch or the validity of its theorems. Applicability is L3. The contribution score in the abstract is a separate assessment of the deductions proved in this report.

Put $\lambda = \pi/8$, $\sigma = 3/8$, and

$$x_* = (2\cos\lambda)^{-1} = (2+\sqrt{2})^{-1/2}.$$

The dual triangular tiling has side one and honeycomb edges have length $1/\sqrt{3}$. Ports are edge midpoints. A walk receives one factor x_* per distinct visited vertex; terminal half-edges have no additional weight. A strip of h layers has physical height $h\sqrt{3}/2$. Fix one bottom port, sum over all top ports and all finite self-avoiding bridge lengths, and write

$$\mu_h(\gamma) = x_*^{L(\gamma)}, \quad b_h = \sum_\gamma \mu_h(\gamma), \quad \ell_h = \sum_\gamma L(\gamma)\mu_h(\gamma), \quad P_h = \mu_h/b_h. \tag{1}$$

Here $L(\gamma)$ is the number of visited vertices and $X(\gamma)$ is the largest horizontal distance of one of them from the source. This is not the uniform fixed-length walk law, a worst-endpoint law, or the law of mixed exits returning to either wall.

The source asserts, with $h \to \infty$ through admissible strip heights,

$$b_h = h^{-1/4+o(1)}, \qquad \ell_h = h^{13/12+o(1)}, \qquad \mathbb{E}_{P_h} L = h^{4/3+o(1)}. \tag{2}$$

These statements, and not merely finiteness at each fixed h, are hypotheses of our asymptotic corridor corollary. In contrast, its nonasymptotic inequalities apply whenever the explicitly stated mass, geometry, and tail bounds are certified.

1.1 Architecture actually read

All 18 cached public text files, including all scientific sections and the three figure sources, were literally displayed in bounded chunks. The architecture is as follows.

1. `model` proves local parafermionic flux, strip balance, and fixed-width cylinder summability. Its finite-width decay constants are not asserted uniform as the circumference grows.

2. `rows` gives the checked matrix, cup and rank-three reductions, Yang–Baxter slides, commuting twisted rows, and a simple physical eigenline. `vacuum` constructs a polynomial representative using refined spin-dependent degrees, then identifies its empty component.

3. `pairing` identifies the normalized bilinear vacuum pairing with a positive polygon polynomial at physical sites. `complex` uses finite residues, a radial regulator, a full-period Fourier estimate, and an analytic pressure correction to prove its claimed bulk bound.

4. `boundary` derives an arch deficit from a row derivative, converts a biorthogonal representer to a positive pole-set mixture, and uses low-pole saturation and weighted polynomial comparisons.

5. `localization` combines a zero-free disk with weighted incompatibility and a six-rotation geometric argument. The physical covering metric is not the auxiliary pillowcase metric.

6. `visit-recursion`, `lateral-tail`, and `visits` convert polygon nesting into visits and mixed-exit means. Finally `bridge` uses a marked tripod, intersecting pairs, and translated bridge contacts to isolate bridges.

Reading all this text does not independently establish every infinite sum, geometric assertion, or asymptotic estimate. Our proved results are explicitly separated below from this imported chain.

1.2 Details that must be retained

The source mixed-exit mean does not supply the lower bridge mean. Its bridge proof has the genuinely additional comparison

$$J \le C(J_{CC} + m), \quad J_{CC} \le h^{3/4+o(1)}m + O(h^{-100}), \quad J \ge h^{3/4-o(1)}u.$$

Here m, u are once-counted bridge and same-side-arch weights through a bulk vertex, whereas the ordered excursion weight is $2(m+u)$. The fixed-source length mass is exactly the sum of $m(v)$ over one representative of each horizontal vertex orbit, not over all vertices of the infinite strip. This explains both the extra factor of order h and the normalization in (2).

The source also states that a companion manuscript gives the same mean exponent in a fixed Ch corridor. We do not audit that companion or advertise our logarithmic corridor as the strongest available confinement claim. Our result supplies *relative error tending to zero* from the particular tail and first-mass interface in the assigned manuscript.

2 Polynomial identities versus physical normalization

Set $q = e^{i\lambda}$, $t = q^2$, $T = t^3$, $\zeta = t^2 = i$, $n_0 = \sqrt{2} - 1$, and $V = \mathbb{C}^{\{0,+1,-1\}}$. The source cup C and embedding I satisfy

$$C^tC = -n_0, \quad I^tI = n_0\,1_V, \quad C^tI = 0.$$

Here t on a matrix is transpose, not conjugate transpose. Thus

$$P_0 = -CC^t/n_0, \quad P_1 = II^t/n_0, \quad P_2 = 1 - P_0 - P_1$$

are bilinear idempotents. They need not be Hermitian orthogonal projections. The checked matrix is

$$F(z) = a(z)P_0 + b(z)P_1 + a(z)b(z)P_2, \qquad a(z) = \frac{t^2 - z}{t^2 z - 1}, \quad b(z) = \frac{T - z}{Tz - 1}.$$

Its cup and second fusion values are $F(T) = CC^t$ and $F(t^2) = II^t$; these preserve the source signs rather than replacing C^tC by a positive norm.

The polynomial vacuum $\Psi_N(X)$ has total degree $\binom{N}{2}$; each variable has degree at most $N-1$, a $+$ spin reduces the corresponding degree to at most $N-2$, and a $-$ spin forces divisibility by that variable. Fusion values alone are not a replacement for these refinements: the final exchange and cyclicity discrepancies are eliminated using them. In particular, a finite interpolation implementation must enforce the correct spin subspace before counting constraints.

Let $S_N(X)$ be the empty component. The normalized vector is $\Omega_N = \Psi_N/S_N$, only where $S_N \neq 0$. The source physical eigenline justifies this normalization near the homogeneous physical data. Symmetry of S_N is not an immediate empty-component consequence of exchange, since that component mixes with nonempty spins; the source uses physical row geometry and polynomial continuation. For even N, its scalar formula is

$$S_N(X) = \prod_{i<j} \frac{x_i^2 + \sqrt{2}x_i x_j + x_j^2}{x_i - x_j} \operatorname{Pf}\left[\frac{x_i^2 - x_j^2}{x_i^2 + \sqrt{2}x_i x_j + x_j^2}\right]_{i,j}. \tag{3}$$

Coincident sites require the removable polynomial extension, not separate evaluation of the singular factors. This is a source identity, not an arithmetic certificate of a nonzero complex denominator everywhere.

Proposition 2.1 (Scale-covariant normalization error)**.** *Let ℓ be the empty-coordinate functional on a finite normed spin space. Suppose $\ell(u) = S \neq 0$, $\ell(\widehat{u}) = \widehat{S} \neq 0$, and let $e = \widehat{u} - u$, $e_0 = \ell(e)$. Then*

$$\frac{\widehat{u}}{\widehat{S}} - \frac{u}{S} = \frac{e - e_0\widehat{\Omega}}{S}, \qquad \widehat{\Omega} = \widehat{u}/\widehat{S}. \tag{4}$$

Consequently if $|e_0| \leq \delta < |\widehat{S}|$ and $\|e - e_0\widehat{\Omega}\| \leq r$, the error is at most $r/(|\widehat{S}| - \delta)$.

Proof. Multiply the difference by $S = \widehat{S} - e_0$ and cancel $\widehat{S}\widehat{\Omega} = \widehat{u}$. The denominator bound is the reverse triangle inequality. □

The numerator is the error transverse to the selected normalization. For $\widehat{u} = au$, $a \neq 0$, it vanishes exactly, even if $\|\widehat{u} - u\|$ is large. This is a classical quotient identity specialized to the source vacuum, not a new polynomial existence theorem. The required r, δ must come from actual coefficient or interpolation error bounds. Checking exchange at a few unisolvent nodes without degrees, or observing a small residual without a stability constant, does not supply them.

The physical polygon pairing uses

$$\Omega_N(X^{-1})^t D^A_{\kappa/\zeta} \Omega_N(X), \qquad \eta = \kappa + \kappa^{-1};$$

it is bilinear at the algebraic stage. Positive polygon coefficients appear after the source physical orientation and seam cancellation. They are not positivity of a complex transfer matrix.

3 A positive boundary quotient with the original pole law

Fix $N = 2m \geq 2$, put $s = p^2$, and let μ be the pushforward of

$$\frac{\cosh(3\lambda p)\sinh(\lambda p)}{\sinh(8\lambda p)}\, dp \quad (p \in \mathbb{R}),$$

with its continuous value at zero. The test and trial spaces are

$$\mathcal{F}_m = \operatorname{span}\{s^j, s^j p \tanh(\lambda p) : 0 \leq j < m\}, \quad \mathcal{G}_m = \operatorname{span}\{s^j, s^j p \coth(\lambda p) : 0 \leq j < m\}.$$

They are distinct N-dimensional spaces. The source solution $R \in \mathcal{G}_m$ satisfies

$$\int [-R(s)] f(s)\, d\mu(s) = f(0) \quad (f \in \mathcal{F}_m).$$

Its path identification is the exact source normalization

$$d_N := \frac{1}{2\cos(3\lambda)} - G_N = \frac{\sin\lambda}{N}\,[-R(-1)]. \tag{5}$$

G_N counts rightward half-plane arches of displacement less than N whose cylinder projection is simple. It is not a finite-path-length truncation.

The actual pole sets are

$$\mathcal{A} = \{(8j)^2 : j = \tfrac{1}{2}, \tfrac{3}{2}, \ldots\}, \qquad \mathcal{B} = \{(8j)^2 : j = 1, 2, \ldots\}.$$

For $\theta = (A, B)$, $|A| = |B| = m$, define

$$\pi_A(s) = \prod_{a\in A}(1 + s/a), \quad w_\theta = \frac{\mu}{\pi_A \pi_B}, \quad v_A = \Delta(A)^2 \prod_{a\in A} a^{1-N}, \quad v_\theta = v_A v_B.$$

For any positive measure w with the indicated finite moments, write

$$Z_j(w) = \frac{1}{j!}\int \Delta(s_1, \ldots, s_j)^2 \prod_i dw(s_i), \qquad Z_0 = 1.$$

The source proves $0 < \mathcal{B}_N := \sum_\theta v_\theta Z_N(w_\theta) < \infty$ and

$$-R(-1) = \frac{1}{\mathcal{B}_N}\sum_\theta v_\theta Z_N(w_\theta)\frac{K_N^{w_\theta}(0,-1)}{\pi_B(-1)}, \qquad \frac{\sin\lambda}{\lambda} \le \pi_B(-1) \le 1. \tag{6}$$

The pole choices are correlated through $Z_N(w_\theta)$; replacing them by independent, uniformly chosen, or most likely poles changes the law.

3.1 The two-external-node identity

Lemma 3.1 (Positive Christoffel numerator)**.** *Let w be a positive measure on $[0,\infty)$ with moments through degree $2N-2$, whose moment matrix on degrees $< N$ is positive definite. For $u \ge 0$,*

$$Z_N(w)K_N^w(0,-u) = Z_{N-1}(s(s+u)w). \tag{7}$$

In particular this kernel value is positive for $u > 0$ whenever the right-hand determinant is positive, as it is for the continuous source weights.

Proof. Let $v(s) = (1, s, \ldots, s^{N-1})^t$ and $G = \int v(s)v(s)^t\,dw(s)$. Expansion of the determinant and permutation integration gives $Z_N(w) = \det G$. The reproducing kernel is $v(x)^t G^{-1} v(y)$. Expanding its cofactor numerator and applying the same determinant integration to the $N-1$ remaining columns gives

$$\det G\, K_N^w(x,y) = \frac{1}{(N-1)!}\int \Delta(s)^2 \prod_{i=1}^{N-1}(x - s_i)(y - s_i)\,dw(s_i).$$

At $x = 0, y = -u$ each factor becomes $s_i(s_i + u)$. All integrals exist under the stated moment hypothesis, since each variable has degree at most $2N-2$. This also proves the formula for atomic measures; no smooth density is needed. □

This is a classical Andréief/Christoffel identity. The source already uses its cofactor version. Our additional use is the explicit positive error accounting below, not a claim to have discovered the identity.

Theorem 3.2 (Boundary certificate as one positive ratio)**.** *Assume the source pole representation* (6)*. Then*

$$-R(-1) = \frac{\mathcal{A}_N}{\mathcal{B}_N}, \qquad \mathcal{A}_N = \sum_\theta \frac{v_\theta}{\pi_B(-1)} Z_{N-1}(s(s+1)w_\theta). \tag{8}$$

Every summand is nonnegative. Thus finite retained pole sets and positive integration domains give increasing lower approximations separately to $\mathcal{A}_N, \mathcal{B}_N$, but their ratio need not be monotone.

Proof. Substitute (7) in each summand of (6). All factors are positive, so Tonelli's theorem permits the resulting summation and integration without cancellation. The source finiteness and the uniform envelope proved next ensure finiteness of the numerator. Ratios of two increasing positive sequences need not increase, as the example below confirms. □

3.2 A finite common envelope, with no asymptotic constants

Completing both positive pole products gives $w_\theta \geq w_\infty$, where w_∞ is the pushforward of

$$f_\infty(p)\,dp, \qquad f_\infty(p) = \frac{\lambda p(2\cosh(2\lambda p) - 1)}{\sinh(8\lambda p)}. \tag{9}$$

All inequalities here are inequalities of positive measures on a real domain. No analogous order is claimed on a complex contour.

Proposition 3.3 (Uniform finite envelope)**.** *Let $H_N = (\int_1^2 s^{i+j}\,ds)_{0\leq i,j<N}$, and put*

$$a_N = v(0)^t H_N^{-1} v(0), \quad b_N = v(-1)^t H_N^{-1} v(-1), \quad U_N = 648(a_N + b_N).$$

Then $H_N > 0$, its entries and U_N are rational, and for every actual source pole pair

$$0 < \frac{K_N^{w_\theta}(0,-1)}{\pi_B(-1)} \leq U_N. \tag{10}$$

A sharper optional envelope is

$$\frac{\lambda}{\sin\lambda}\sqrt{K_N^{w_\infty}(0,0)K_N^{w_\infty}(-1,-1)}.$$

Proof. After the two-to-one change $s = p^2$, the density of w_∞ is $\lambda(2\cosh(2\lambda\sqrt{s}) - 1)/\sinh(8\lambda\sqrt{s})$. On $1 \leq s \leq 2$ it is at least $\lambda e^{-8\lambda\sqrt{2}} > 1/648$: use $\lambda > 3/8$, $\pi\sqrt{2} < 5$, and $e < 3$. Thus the source moment matrices satisfy

$$G_\theta \succeq G_\infty \succeq H_N/648 > 0.$$

Positive definiteness of H_N follows because a nonzero polynomial cannot vanish almost everywhere on an interval. Inversion reverses the positive-definite order, as follows by conjugating by a square root and diagonalizing. Hence each diagonal kernel is bounded by the corresponding $648\,v^t H_N^{-1} v$. Cauchy–Schwarz in G_θ^{-1} bounds the absolute off-diagonal kernel by the geometric mean. Its sign is supplied by Lemma 3.1, not by matrix order. Finally $\lambda/\sin\lambda < 2$, so $2 \cdot 648\sqrt{a_N b_N} \leq 648(a_N + b_N)$. Retaining G_∞ instead gives the optional bound. □

This envelope is deliberately conservative and may grow rapidly with N. It is useful because it is an entirely specified finite certificate; the source sharper $C\sqrt{\log(2N)}N^{3/4}$ has an unspecified constant. There is no efficiency claim for either exact inversion or pole enumeration.

For completeness, moments for the optional envelope have the explicit positive-term series

$$\int s^k \, dw_\infty(s) = \frac{4(2k+1)!}{\lambda^{2k+1}} \sum_{r\geq 0} \left\{ \frac{1}{(16r+6)^{2k+2}} + \frac{1}{(16r+10)^{2k+2}} - \frac{1}{(16r+8)^{2k+2}} \right\}. \tag{11}$$

To prove it, use $1/\sinh(8\lambda p) = 2\sum_{r\geq 0} e^{-(16r+8)\lambda p}$ for $p > 0$, combine the three terms at each r, and integrate $\int_0^\infty p^{2k+1} e^{-ap}\, dp = (2k+1)!/a^{2k+2}$. Each combined term is positive since its first summand exceeds the subtracted one. Monotone convergence therefore justifies the grouped expansion. For $q = 2k+2 > 1$, the tail after indices $0, \ldots, J-1$ is bounded above by the sum, for $d = 6, 10$, of

$$(16J+d)^{-q} + \frac{(16J+d)^{1-q}}{16(q-1)}.$$

This follows by dropping the negative summand and applying the decreasing integral test. Certified intervals for λ, moment tails, and the inverse Gram matrix would turn the sharper option into arithmetic data; a floating-point inverse alone would not.

3.3 Omitted poles and integration error

Theorem 3.4 (Original-law truncation ledger)**.** *Let Θ_0 be a finite nonempty retained set of actual pole pairs. Use the same summands as in (8), and denote retained totals by $A_0, B_0 > 0$. Suppose $0 \leq B_{\text{out}} \leq eB_0$, $e \geq 0$. Then with $r_0 = A_0/B_0$,*

$$\frac{r_0}{1+e} \leq -R(-1) \leq \frac{r_0 + eU_N}{1+e}. \tag{12}$$

More generally, if certified integrations give $A_- \leq A_0 \leq A_+$, $0 < B_- \leq B_0 \leq B_+$, and $B_{\text{out}} \leq T$, a directly usable interval is

$$\frac{A_-}{B_+ + T} \leq -R(-1) \leq \frac{A_+ + U_N T}{B_-}. \tag{13}$$

Multiplication by $\sin\lambda/N$ gives an interval for the physical arch deficit in (5).

Proof. By (10), every summand ratio belongs to $[0, U_N]$. The omitted numerator satisfies $0 \leq A_{\text{out}} \leq U_N B_{\text{out}}$. Writing $x = B_{\text{out}}/B_0$ gives the lower bound $r_0/(1+x)$ and upper bound $(r_0 + xU_N)/(1+x)$. The former decreases with x; the latter increases since $r_0 \leq U_N$. Use $x \leq e$. The looser interval with integration errors follows by separate positive numerator and denominator bounds. □

The missing condition $B_{\text{out}} \leq T$ is a real obligation. The source low-pole saturation theorem is an asymptotic probability statement with hidden constants; it is not automatically a numerical certificate for T at a chosen N. Positive finite terms provide lower bounds, never an upper bound on an uninspected infinite remainder. Formula (13) permits any rigorously proved omitted weight bound without replacing the pole law by a sampled approximation.

Proposition 3.5 (Correlated scalar and shape errors)**.** *Fix N, and use one positive common measure in all pole weights. Suppose its approximation is $\widehat{\mu} = a(1+e(s))\mu$, with $a > 0$ and $|e(s)| \leq \varepsilon < 1$ almost everywhere. For either the full positive sums, or one fixed common retained pole set, let r and $\widehat{r}$ be the ratios in (8). Then*

$$\frac{1}{a}\frac{(1-\varepsilon)^{N-1}}{(1+\varepsilon)^N} \leq \frac{\widehat{r}}{r} \leq \frac{1}{a}\frac{(1+\varepsilon)^{N-1}}{(1-\varepsilon)^N}. \tag{14}$$

For a pure scalar error ($\varepsilon = 0$), the factor is exactly a^{-1}.

Proof. The numerator contains $N-1$ integrations and the denominator N. Positivity bounds each product of density factors between the corresponding powers. The common factors a^{N-1} and a^N cancel to a^{-1}. Sum the positive bounds before dividing. □

For a total retained numerator relative error $\alpha < 1$ and denominator relative error $\beta < 1$, an alternative elementary interval is $(1-\alpha)/(1+\beta) \le \widehat{r}/r \le (1+\alpha)/(1-\beta)$. Proposition 3.5 is preferable when a large scalar rescaling is known exactly. An arbitrary small entrywise moment discrepancy is not a substitute for either hypothesis.

Example 3.6 (Exact low-dimensional fixture)**.** For $N=2$, take the test measure $w = \delta_1 + \delta_2$, not the continuous physical source measure. Then

$$G = \begin{pmatrix} 2 & 3 \\ 3 & 5 \end{pmatrix}, \quad G^{-1} = \begin{pmatrix} 5 & -3 \\ -3 & 2 \end{pmatrix}, \quad Z_2 = 1, \quad K_2(0,-1) = 8, \quad Z_1(s(s+1)w) = 8.$$

With the genuine pole value $B = \{64\}$, the kernel factor is $8/(1-1/64) = 512/63$. This checks the sign and external-node denominator, but it is not a computed honeycomb amplitude. For another normalization fixture combine w and $2w$ with equal outside coefficients and external denominator one. Their denominator weights are $1, 4$, numerator weights $8, 16$, so the full ratio is $24/5$, below the first retained ratio 8. This demonstrates why monotone positive truncation cannot be asserted for the ratio.

4 Finite corridor certificates for the actual bridge law

The source lateral-tail lemma gives absolute $C_0, c > 0$, uniformly in the strip height and convex lateral truncations, such that

$$\mu_h(X \ge t) \le C_0 e^{-ct/(h+1)} \quad (t \ge 0). \tag{15}$$

The geometric counting bound is

$$L \le C_g(h+1)(X+1). \tag{16}$$

It uses self-avoidance and the number of vertices in a strip box. It would be false for arbitrary walks allowed to revisit vertices. Both bounds apply to the positive bridge submeasure of the source exit measure. We retain the actual weights (1) throughout.

Let $H = h+1$ and retain $E_t = \{X < t\}$. Define

$$B_t = \int_{E_t} d\mu_h, \quad L_t = \int_{E_t} L\, d\mu_h, \quad D_0(t) = b_h - B_t, \quad D_1(t) = \ell_h - L_t.$$

An equality-at-the-cut convention can be changed by replacing t with an adjacent admissible value; the tail inequalities below use the declared closed discarded event.

Theorem 4.1 (All fixed insertion tails)**.** *For every integer $j \ge 1$, under (15)–(16),*

$$\begin{aligned} D_j(t) &:= \int_{X \ge t} L^j \, d\mu_h \\ &\le C_0 C_g^j H^j e^{-ct/H} \left[(t+1)^j + \sum_{k=0}^{j-1} \frac{j!}{(j-1-k)!} (t+1)^{j-1-k} \left(\frac{H}{c}\right)^{k+1} \right]. \end{aligned} \tag{17}$$

Also $D_0(t) \le C_0 e^{-ct/H}$. In particular,

$$D_1(t) \le C_0 C_g H (t + 1 + H/c) e^{-ct/H}. \tag{18}$$

These are upper bounds, not sharp asymptotic formulas for higher moments.

Proof. For $X \geq t$, the layer-cake identity gives

$$\int_{X\geq t} (X+1)^j \, d\mu_h = (t+1)^j \mu_h(X \geq t) + j \int_t^\infty (u+1)^{j-1} \mu_h(X > u) \, du.$$

Tonelli applies to nonnegative integrands, including atomic laws. Insert the exponential tail. With $u = t+v$, expand $(t+1+v)^{j-1}$; each integral of $v^k e^{-cv/H}$ equals $k!(H/c)^{k+1}$, by repeated integration by parts. Multiply by $C_g^j H^j$. The case $j = 0$ is exactly the tail hypothesis. □

Theorem 4.2 (Normalization and the two changes of law)**.** *Suppose $b_h, \ell_h > 0$, $D_0 < b_h$, and $D_1 < \ell_h$. Put $\epsilon_0 = D_0/b_h$, $\epsilon_1 = D_1/\ell_h$. For the ordinary law P_h, the length-biased law $Q_h(\gamma) = L(\gamma)\mu_h(\gamma)/\ell_h$, and their conditional laws $P_{h,t}, Q_{h,t}$ on E_t,*

$$\|P_h - P_{h,t}\|_{\mathrm{TV}} = \epsilon_0, \qquad \|Q_h - Q_{h,t}\|_{\mathrm{TV}} = \epsilon_1, \tag{19}$$

$$\frac{\mathbb{E}_{P_{h,t}} L}{\mathbb{E}_{P_h} L} = \frac{1-\epsilon_1}{1-\epsilon_0}, \qquad \left| \frac{\mathbb{E}_{P_{h,t}} L}{\mathbb{E}_{P_h} L} - 1 \right| \leq \frac{\max(\epsilon_0, \epsilon_1)}{1-\epsilon_0}. \tag{20}$$

Here total variation is the supremum over measurable events, not twice that supremum. Furthermore the Wasserstein distance of the two length laws, with cost $|x-y|$, is bounded by

$$W_1(\mathcal{L}_{P_h}(L), \mathcal{L}_{P_{h,t}}(L)) \leq \frac{D_1}{b_h} + \frac{D_0}{b_h}\frac{L_t}{B_t}. \tag{21}$$

Proof. Any probability P decomposes into $(1-\epsilon)P(\cdot|E) + \epsilon P(\cdot|E^c)$; the two components have disjoint supports, giving the total-variation equality. Apply this first to P_h, then to Q_h. The mean quotient follows from L_t/B_t. Its error numerator is $|\epsilon_0 - \epsilon_1|$, at most their maximum. For (21), couple retained samples identically and, on the discarded component, draw an independent retained length. The expected cost is at most the sum of the discarded length contribution D_1/b_h and $\epsilon_0 L_t/B_t$, since lengths are nonnegative. □

This separates two errors that ordinary total variation alone cannot control. A tiny discarded bridge mass may carry most of the first length moment.

4.1 A usable finite-height interval

Suppose a finite corridor computation gives exact positive B_t, L_t and certified tail bounds $D_0 \leq d_0, D_1 \leq d_1$. Then

$$\frac{L_t}{B_t + d_0} \leq \mathbb{E}_{P_h} L \leq \frac{L_t + d_1}{B_t}. \tag{22}$$

Positive interval enclosures for B_t, L_t can be inserted in the appropriate denominator and numerator directions. This is an actual original-law finite certificate if the constants C_0, c, C_g and the retained sums are certified. The source only states the existence of its uniform constants; this report does not invent their numerical values. Finite-state enumeration in a corridor is finite because self-avoidance bounds the length by its number of vertices, but its running time is not asserted practical.

5 Logarithmic corridors and the optimal hypothesis-level threshold

Theorem 5.1 (Relative preservation of fixed moments)**.** *Assume the source powers (2) and the uniform tail and geometric hypotheses (15)–(16). Fix an integer $j \geq 1$, let*

$$M_{j,h} = \int L^j \, d\mu_h, \qquad t_h = A(h+1)\log(h+1), \qquad A > 0, \qquad s_j = \frac{2j}{3} + \frac{1}{4}.$$

All $M_{j,h}$ are finite. If $cA > s_j$, then

$$\frac{\mathbb{E}_{P_{h,t_h}} L^j}{\mathbb{E}_{P_h} L^j} \longrightarrow 1, \qquad \|Q_{j,h} - Q_{j,h}(\,\cdot\,|E_{t_h})\|_{\mathrm{TV}} \longrightarrow 0, \tag{23}$$

where $Q_{j,h} = L^j \mu_h / M_{j,h}$. More precisely,

$$\epsilon_{0,h} := D_0(t_h)/b_h \le h^{1/4-cA+o(1)}, \tag{24}$$

$$\epsilon_{j,h} := D_j(t_h)/M_{j,h} \le h^{s_j-cA+o(1)} (\log h)^j. \tag{25}$$

For any fixed $\delta > 0$, choosing $cA = s_j + 2\delta$ gives $\epsilon_{j,h} \le h^{-\delta}$ eventually. For $j = 1$, the cutoff condition is $cA > 11/12$, and (21) divided by $\mathbb{E}_{P_h} L$ tends to zero as well.

Proof. Finiteness follows from (17) at $t = 0$. Jensen's inequality under the probability P_h gives

$$M_{j,h} \ge b_h (\ell_h/b_h)^j = \ell_h^j / b_h^{j-1} = h^{4j/3-1/4+o(1)}. \tag{26}$$

This is a lower bound only; we do not identify the higher-moment exponent. For fixed j, A, c, every term in the bracket of (17) is $O(H^j (\log H)^j)$. Thus

$$D_j(t_h) \le C_{j,A,c} H^{2j-cA} (\log H)^j, \qquad D_0(t_h) \le C_0 H^{-cA}.$$

Divide by (26) and $b_h = h^{-1/4+o(1)}$. Since $H = h + 1$, replacing H by h changes only harmless factors. For $cA = s_j + 2\delta$, the $o(1)$ and the fixed power of the logarithm are eventually absorbed by h^δ.

The conditioned j-th-moment ratio is exactly $(1 - \epsilon_{j,h})/(1 - \epsilon_{0,h})$, and the length-$j$-biased total variation is $\epsilon_{j,h}$, by the same disjoint-mixture proof as Theorem 4.2. These converge as claimed. For $j = 1$, divide (21) by ℓ_h/b_h; the resulting bound is $\epsilon_{1,h} + \epsilon_{0,h}(1 - \epsilon_{1,h})/(1 - \epsilon_{0,h})$. □

The quantifiers are important: j, A, δ are fixed before $h \to \infty$. A single fixed A does not satisfy these sufficient conditions for all unbounded j. The unspecified source c and the thresholds implicit in its $o(1)$ estimates prevent an effective numerical height cutoff here. A stronger finite lower bound for $M_{j,h}$ can be substituted directly and may improve the sufficient corridor.

Example 5.2 (A rate budget, not a measured constant)**.** If one supplies a valid value of the source decay constant c, take $A = (11/12 + 1/5)/c$. Then for the first moment

$$\epsilon_{1,h} \le h^{-1/10} \quad \text{eventually}, \qquad \epsilon_{0,h} \le h^{-13/15+o(1)}.$$

The exponent $13/15 = 2/3 + 1/5$ concerns ordinary mass, and is not the first-moment error exponent. The relative mean is the exact ratio in (20), so it also converges to one. This illustrates how to allocate an exponent budget; it does not assign a numerical value to c.

Theorem 5.3 (Sharpness for the declared interface)**.** *Fix $j \ge 1$, $c > 0$, and $A > 0$ with $cA \le s_j$. There exist positive two-atom measures, with nonnegative observables called X, L, satisfying (15), (16), and both mass exponents in (2), for which*

$$\frac{\mathbb{E}_{P_{h,t_h}} L^j}{\mathbb{E}_{P_h} L^j} \longrightarrow \frac{1}{2}.$$

The measures are abstract fixtures, not realizations by honeycomb walks. Consequently the threshold s_j cannot be lowered, or made non-strict, using only these hypotheses.

Proof. Write $H = h + 1$, and for all sufficiently large H use the two atoms

	weight	X	L
retained	$H^{-1/4}$	$H^{1/3}$	$H^{4/3}$
discarded	$H^{-s_j}(\log H)^{-j}$	$AH\log H$	$H^2\log H.$

The retained atom lies below $t_h = AH\log H$, whereas equality places the second atom in the discarded event. The volume bound holds with $C_g = \max(1, A^{-1})$. Total mass is $H^{-1/4}(1 + o(1))$. The retained first mass is $H^{13/12}$; the discarded first mass is

$$H^{7/4-2j/3}(\log H)^{1-j},$$

which has no larger power, with equality when $j = 1$. Thus the first mass is $H^{13/12+o(1)}$.

For $0 \le t \le H^{1/3}$, total mass is at most $3e^{-ct/H}$ for all large H, because $t/H \le H^{-2/3}$. For $H^{1/3} < t \le AH\log H$, only the second atom remains and its weight is at most H^{-cA}, since $cA \le s_j$ and $\log H \ge 1$. For larger t the tail is zero. Hence (15) holds with $C_0 = 3$, uniformly for this family after a fixed initial height. Finitely many initial heights can be omitted or filled in with an enlarged constant.

Each atom contributes exactly $H^{4j/3-1/4}$ to the j-th length mass. Thus $\epsilon_{j,h} = 1/2$, while $\epsilon_{0,h} \to 0$. The exact normalization identity gives the limit $1/2$. The displayed L's need not be integer walk lengths; rounding them up and adjusting C_g gives the same asymptotic counterexample if integer-valued observables are desired. □

For $j > 1$ this counterexample can have asymptotically unchanged first mean while failing to preserve the j-th moment. Therefore a successful first-moment confinement test cannot be silently promoted to all length-biased laws.

6 Complex representation: retained obligations and finite limits

The source bulk quantity is the degree-at-most-n positive polynomial $F_n(\eta)$ for disjoint polygons separating the two marks of a balanced cylinder of circumference $2n$. The source estimate is

$$|F_n(2\cosh s)| \le C_K n^{1/6+2\,\mathrm{Re}(s^2)/(3\pi^2)} \quad (s \in K \Subset \{|\operatorname{Im} s| < \pi\}).$$

The exponent here involves $\mathrm{Re}(s^2)$, not $|s|^2$. Neither the finite boundary quotient nor the corridor results prove this estimate.

Several details of the actual argument obstruct a shortcut from finite interpolation to that asymptotic claim.

1. The finite neutral-color representation has activities satisfying $\lambda_+\lambda_- = \sqrt{2} - 2 < 0$. Before the physical regrouping, it is not a probability distribution. The virtual counts bounded by $2n$ are not walk lengths.

2. The confluent Pfaffian denominator is an even moment determinant with a positive real weight. Its comparison with a reference orthogonal ensemble must retain the stated $n^{5/48}$ factor and bounded logarithmic error; finite Gram positivity alone supplies neither.

3. The positive full-period Fourier estimate includes the regular zero mode, self-interactions, and a linear phase from the antisymmetric regular part. Dropping zero or retaining only pairwise off-diagonal terms changes the quadratic form. A matrix-valued positive Fourier form is not entrywise positivity.

4. The source compares a small-slope complex contour to a real contour with coercive box-population bounds. It does not declare a quadratic form positive on an arbitrary complex path.

5. At period $L = (3/2)\log n + i\theta$, the pressure correction $c_* n^2 e^{-L}$ is of order $n^{1/2}$, not $O(1)$. The analytic factor removing it is retained before Cauchy's formula recovers the unregularized value.

The source then uses the special normalized value at $s = i\pi/2$, analytic compactness, and positive polynomial coefficients to obtain the claimed zero-free disk and fugacity-two exponent. Those hypotheses are not checked by our atomic moment fixtures. Likewise, the six-rotation argument and the tripod transfer remain genuine geometric input. No mathematical source correction is asserted on the basis of this scoped analysis.

7 Original Ram comparison and practical tradeoffs

Fresh original proof reading was local and bounded. In the original TR250, lines 155–180 define the actual finite measurement map $m = Ec$, distinguish compatible data from arbitrary off-range data, and exclude an infinite-dimensional lower frame conclusion. Lines 212–258 prove compatible coordinate transfer with the physical mass matrix. Lines 455–476 prove polynomial moment unisolvence and explain why too few positive quadrature nodes do not preserve it. The separately read residual proof at lines 478–594 retains the actual weak load and boundary obligations; we use its error-accounting principle, not a diffusion theorem about honeycomb walks [7].

The original RamCS chapter, freshly read through line 213, specifies weighted Fourier inversion and Plancherel, differentiability under a majorant, convolution signs, and the difference between a local symbol and an operator eigenfunction [8]. Its included companion files were not thereby read. In the source complex representation, absolute summability or a specified L^2 limiting argument is still required before rearranging Fourier integrals; analytic continuation does not preserve positive real-measure order.

The physical-metric chapter was freshly read in full. For synthesis $Jc = \sum c_i \phi_i$, its norm is $c^* G c$, $G = J^* J$, not necessarily $\|c\|_2^2$. Its full residual identity includes the omitted orthogonal component [9]. In our boundary construction the moment matrix is exactly that physical Gram matrix. If $v = R\widetilde{v}$ is an invertible change of polynomial coordinates, both the Gram matrix and evaluation vectors must transform. The scalar $v(x)^t G^{-1} v(y)$ is unchanged, as follows by cancellation of R, R^{-1}. This coordinate fact is classical, not the new contribution claimed here.

Advantages. The two-external-node formula preserves the original pole distribution and makes all retained contributions positive. The explicit rational envelope offers a finite upper certificate without using an unknown asymptotic comparison constant. Scalar normalization errors can be removed exactly. The corridor theorem supplies a conditional polynomial error budget for the actual bridge law and its fixed length-biased laws, not just equality of their logarithmic mean exponents.

Costs and limitations. The rational envelope can be very large and positive partition sums can require many pole choices. The integration dimension grows with N. No method here supplies an uncomputed omitted-pole bound for free. The source tail constant is not numerically extracted. A finite corridor enumeration may be infeasible despite being finite. The abstract sharpness example limits these hypotheses, not the true lattice model: additional geometric information can yield narrower corridors.

Plausible extensions with missing inputs. A sharper certified lower measure than the interval minorant in Proposition 3.3 can reduce the kernel envelope. Formula (11) offers one route, but requires rigorous moment intervals and a verified inverse margin. A quantitative low-pole-saturation proof would supply effective B_{out}; the source qualitative statement alone does not. Noninteger insertion powers can be treated by an incomplete-gamma tail integral, but a uniform range of growing powers needs new quantitative constants. Improving $Ah\log h$ to fixed Ch while preserving the mean *relatively* needs information beyond (15)–(2); equality of mean exponents in a companion paper is not itself that assertion.

8 Literature status and audit scope

The October 9, 2026 primary-source refresh distinguishes several results. Duminil-Copin and Smirnov's published connective-constant theorem fixes the critical activity, not the source mean exponent [2]. The arXiv abstract of Krachun and Panagiotis states quantitative sub-ballisticity for the length-n law and polynomial decay of critical bridge partition functions; this is a different assertion and ensemble [3]. Garbali and Nienhuis study the dilute $O(1)$ model with open boundaries, including ground-state recurrences and sum-rule determinants [4, 5]. Those primary abstract/version pages support context, not direct transfer of their theorems to the source periodic $n = 0$ vacuum. Their complete proofs were not freshly audited here.

The publisher's current formalization-scope page explicitly places the assigned manuscript's mean and visit exponents outside the listed supporting bridge-finiteness formalization [6]. We read the scope description, not the Lean proof files and dependencies. No whole-repository formal-verification claim is made.

The owned read ledger records all 18 public texts, their exact intervals, pre-read and post-read SHA-256 hashes, and the private unread complements. Private reading is 266 lines of TR250, 213 lines of the RamCS chapter, and 210 lines of the physical-metric chapter. All other private text, including referenced inputs and the broader corpus, is outside the fresh read scope. Public searches used only public titles and URLs; no confidential corpus was uploaded.

The independently executable owned checks use rational arithmetic for positive atomic Gram/cofactor identities, actual pole denominators in finite fixtures, mixture truncation and rescaling, layer-cake polynomial identities, normalization, and threshold arithmetic: 2,463 exact-rational cases with 6,660 assertions passed. Four additional 65-digit numerical quadrature/series comparisons checked the corrected $16r$ spacing in (11); these are diagnostics, not interval proofs. The total is 2,467 cases and 6,664 assertions. Counts and script hashes are recorded in `M461-tests.json`; these fixtures are not continuous physical-source amplitudes and do not establish the source bulk or bridge exponents. Parent checks and foreign checkers are excluded. The contribution is the explicit source-linked certificate interface and the conditional sharp moment-transfer result, with classical tools credited. No numerical critical amplitude or measured computational speedup is reported.

References

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[3] D. Krachun and C. Panagiotis, *Quantitative sub-ballisticity of self-avoiding walk on the hexagonal lattice*, arXiv:2310.17299v1 (2023). Primary abstract/version page freshly checked; source bibliography lists *Annals of Probability* 54 (2026), 1109–1125. https://arxiv.org/abs/2310.17299.

[4] A. Garbali and B. Nienhuis, *The dilute Temperley–Lieb $O(n = 1)$ loop model on a semi infinite strip: the ground state*, arXiv:1411.7020v2, January 3, 2017. Primary abstract/version page checked. https://arxiv.org/abs/1411.7020v2.

[5] A. Garbali and B. Nienhuis, *The dilute Temperley–Lieb $O(n = 1)$ loop model on a semi infinite strip: the sum rule*, arXiv:1411.7160v2, January 3, 2017. Primary abstract/version page checked. https://arxiv.org/abs/1411.7160v2.

[6] OpenAI, *The three-quarter exponent for honeycomb self-avoiding walk*, formalization scope description, accessed October 9, 2026. https://github.com/openai/math/blob/main/lean/docs/237.md.

[7] M. Subbarao, original confidential TR250, local file `RamFoundationTRs/TR-rev-1-old-tr250/TR-rev-1-old-tr250.tex`. Fresh literal proof scope: lines 145–270 and 455–594.

[8] M. Subbarao, original *RamCS Transforms and Operator Spectra*, `RamCalBooks`, volume 1 revised, chapter 7, `ch07-ramcs-rev2.tex`. Fresh scope: lines 1–213; included companion files not read.

[9] M. Subbarao, original physical-metric chapter, `RamCalBooks`, volume 21, chapter 1, `ch01d-physical-metrics.tex`. Fresh scope: complete 210-line file.